



Feedback on MESMERICE Sustainability Standards for Crypto Assets

Index

Re: Feedback on MESMERICE Sustainability Standards for Crypto Assets	Page 2
Energy Use as a Policy Objective	Page 3
Environmental Metrics Adopted and Reporting Costs	Page 4
Reliance on Per Transaction Intensity Metrics	Page 5
Technology Design and Protocol Differentiation	Page 6
Comparisons Against Countries	Page 7
Carbon footprint estimations	Page 8
A proper understanding of the technology	Page 9

MiCA Crypto Alliance
15 June 2025

Trinomics
Mauritsweg 44
3012 JV Rotterdam
Netherlands

Re: Feedback on MESMERICE Sustainability Standards for Crypto Assets

Dear Trinomics team,

On behalf of the MiCA Crypto Alliance, I am writing to provide feedback on the MESMERICE Study and the recent webinar "Study for developing a methodology and sustainability standards for mitigating the environmental impact of crypto-assets", conducted as a part of the tender award under FISMA/2023/OP/0005.

We appreciate the effort to strengthen the methodology for assessing the environmental performance of crypto-assets, and welcome the opportunity to contribute constructively to the study's ongoing development. We thank you for interviewing the MiCA Crypto Alliance as a part of the key informant interviews conducted during your research, as well as for considering our comments on a live setting during a recent webinar with other stakeholders.

We would like to share further feedback and highlight a number of concerns for your consideration. These can be summarised as follows:

- **Energy use should not be treated as intrinsically harmful.**
- We recommend **avoidance of highly indirect environmental metrics** from generic LCA models.
- **Land use should be treated as an absolute metric**, not a per-transaction metric.
- **Per-transaction metrics more broadly are problematic.** They fail to capture the purpose of energy use in blockchains, ignore Layer-2 scaling, and conflate micro- and high-value transactions.
- **Protocol design and energy use must be decoupled.** Not all PoW chains are equally intensive, and some PoS systems consume as much or more.
- **Country-level comparisons are misleading.** Industry-to-industry benchmarks (e.g. Bitcoin vs gold or banking) are more appropriate.
- Trinomics' **carbon estimates appear inflated** due to neglect of off-grid mining and layer-2 (L2) usage.
- **Technical misunderstandings must be corrected.** Bitcoin was wrongly implicated in Spanish energy outages. Statements conflating mining with asset creation reflect conceptual inaccuracies.

Energy Use as a Policy Objective

We appreciate the research team's effort to accommodate different viewpoints in their research, and the difficulty that this entails. Nevertheless, we are concerned that the standard emerging from this study might construe energy use as intrinsically harmful, and therefore suggest that energy use reductions constitute an intrinsic policy goal. In our view, this framing could be problematic. While energy use in itself does carry some intrinsic costs, such as grid congestion in the context of connection issues or energy scarcity, in general terms policy priorities should be focussed around carbon footprint, electronic waste, or problematic forms of water use. Crypto asset energy use from off-grid renewable sources is a growing trend and it can also be associated with node decongestion and the development of additional energy sources. Increasing energy use with a stabilisation or even a reduction of carbon footprint, is expected by 2030 and beyond, and this should largely be regarded as a roughly positive development, especially if this study seeks to adopt standard views in the field. We are concerned that the theoretical framework adopted in this study could distort such a position despite many researchers in the field would agree with it.

We have noted that at least some of the researchers in the team make public statements that adopt a “planetary boundaries” framework, a paradigm emergent from the Club of Rome, indirect successor to the “limits to growth” perspective, popularised by the “doughnut economics” model. We see the adoption of this theoretical framework as potentially driving the team's approach to energy use.

While the planetary boundaries framework has stimulated some valuable debate about ecological limits, its operational utility remains limited – as does its adoption. In our opinion, regulation should be guided by more reliable models such as Integrated Assessment Models (IAMs), which though imperfect, offer a more grounded basis for decision-making and allow for better thinking in terms of policy-relevant trajectories. In other words, while we think the planetary boundaries framework may have valuable contributions, it should not be the primary analytical framework guiding EC research with regulatory impact.

To conclude this section, we would like to highlight that energy consumption disclosures are mostly important for reasons: (a) they are a scientific prerequisite for other disclosures such as carbon footprint, water footprint, land use, etc.; [b] they are lower-cost to report on than the latter. Reporting on energy consumption enables a disclosure framework whereby high energy usage is used as a lower-cost “red flag” that triggers other, more sophisticated, indicators. This should not be misconstrued to position energy use as ‘the’ decisive indicator on environmental impact.

Environmental Metrics Adopted and Reporting Costs

Another concern relates to the use of highly indirect environmental metrics such as ionising radiation, photochemical ozone formation, and particulate matter emissions in the context of crypto-assets. In our understanding, these were derived through a life-cycle assessment (LCA) extrapolated from a single UK data centre, and then scaled to the global Bitcoin network. We would like to caution against use of this approach as, in our view, it may introduce significant distortions and lacks sufficient basis for guiding proportional policy action.

While LCA tools can provide value in evaluating hardware footprints, their use requires caution, as reliance on broad estimates risks being misapplied or misunderstood. These metrics presented at the recent MESMERICE webinar have been notably absent from the crypto-asset environmental literature until now. We believe this is no coincidence, but rather a result of the fact that the suggested metrics may be too indirect and hard to establish. We favour point-of-generation emission factors instead.

In addition, the European crypto asset environmental reporting regime probably constitutes already the world's most stringent framework of its kind, with 36 indicators established under Commission Delegated Regulation (EU) 2025/422. We suggest not relying on even further indicators, particularly very indirect indicators which rely on excessively rough assumptions, to maintain costs proportionate and the EU competitive.

Finally, we note with pleasure that the team has adopted 'land use' as a proxy for the usage of natural resources. Our team has been using this very same proxy for a long time, and had suggested such an indicator when we were interviewed by the researchers in charge of this project. This suggestion was made with the goal of facilitating comparability and reducing costs by introducing a widely-accepted proxy. However, we noted that, during the webinar, the Trinomics team presented metrics of "land use per transaction", which are clearly derived from deriving a crypto asset's land use by the number of base layer transactions in a year. We would like to qualify usage of this intensity metric.

Unlike other environmental impact metrics such as energy use, carbon footprint, water use or electronic waste, land use is a variable without a clear time dimension, in the sense that, while it is possible to speak of "energy use of a blockchain in a year", it is not equivalent to speaking of "land use of a blockchain in a year". While such a discourse move is possible, the land is not "used up" or "consumed" as happens with other indicators. Assume a blockchain system with a stable power use of 20 MW, with the power sources occupying 100,000 m². This means the system uses 5 m²/kW or, over a year, 175,200,000 kWh with a land use intensity of 0.00057 m²/kWh. The following year, however, while an additional 175,200,000 kWh of energy will be consumed, no additional 100,000 m² of land will be consumed. Rather, the same land will remain in usage. While energy use accumulates, land use does not, but just extends over time.

This has an important implication. Land use intensity figures in m²/kWh/year exist in the literature, making it technically valid to compare annual energy use with transaction counts over the same period. However, extrapolation of land use "per transaction" metrics can be greatly misleading as, while transactions accumulate across years (and so do other impact metrics), land 'consumption' does not. In consequence, reliance on a land usage per transaction intensity metrics can be dangerously misleading, as it can create the impression that, if e.g. Bitcoin has processed 1.2 billion transactions since its inception, $1.2 \text{ billion} \times 20 \text{ m}^2 = 24,000 \text{ km}^2$, falsely creating the impression that Bitcoin has used more land than the entire size of Wales. The correct number would rather be 1,900 km² instead. In short, we encourage reliance on the land use metric as an absolute indicator, but discourage use of this per transaction intensity metric due to its potential for misinformation.

Reliance on Per Transaction Intensity Metrics

More generally, we would like to caution against indiscriminate usage of per transaction intensity metrics, as impact “per transaction” metrics are known to be misleading in themselves.

While any metric is valid within its limitations, these limitations need to be made pristinely apparent and, ideally, mitigated by resorting to other metrics that complement them. Research on any potentially new sustainability standards should explore alternative metrics, which we have not seen in the presented discussions so far.

Some known limitations of per transaction metrics include that:

- In the highest-energy blockchain systems, most energy consumption (90%-99%) is not caused by transactions but by block subsidy- and coin price-incentivised mining.
- In blocks-based systems, per-transaction figures artificially increase if blocks are less full, despite the marginal energy cost of adding new transactions is zero.
- While not necessarily the case, the majority of “per transaction” intensity metrics (Trinomics’ included) do not account for layer-2 (L2) transactions.
- They do not account for the value of the transactions, considering wholesale transactions and microtransactions as equivalent-value outputs.

As a result of this, we encourage usage of significant caveats when relying on per-transaction metrics, and ideally supplementing them with e.g. per-value-settled metrics. This makes special sense in Proof of Work networks, which tend to be the architecture of choice of blockchains with a monetary use case.

Technology Design and Protocol Differentiation

A potential second-order effect of reliance on per-base-layer-transaction metrics could be that this disincentivises L2 scaling solutions, as their throughput is not computed in efficiency metrics. This could be construed as implicit encouragement for the usage of native L1 scaling solutions, such as larger block size, faster block times, or different emission schedules. All of these architecture choices may reduce energy consumption or energy consumption “per transaction” metrics at the base layer level of analysis. Nevertheless, the language used by the Trinomics team has tended to risk putting blockchains with native scaling architectural choices in the exact same basket as second-layer architectural choices: They are all “Proof of Work like Bitcoin”.

We caution against treating consensus algorithms as the only variable behind energy use. Filecoin, a Proof of Stake network, uses a similar amount of energy to EthereumPoW or Dash, and significantly more than Sia and BSV – all Proof of Work networks.

Comparisons Against Countries

We noted that the webinar presented comparisons of cryptocurrencies against entire countries. Such comparisons, while popular in the past, are known to be misleading and tend to be discouraged. This is due to their contextualise energy intensity or service equivalence: countries are a good benchmark to industries. More appropriate comparisons are typically industry-to-industry (e.g. Bitcoin against gold, DeFi against the banking system, and so on).

Carbon footprint estimations

In the webinar, it was indicated that the Trinomics analysis estimated Bitcoin's carbon footprint at around 90 Mt of CO₂ per year, which roughly aligned with CCAF CBECI (live) data but was twice as much as the latest CCAF report. Our data also coincides with CCAF's latest report. This is no minor detail, as CBECI (live) data relies on 2022 miner locations and the assumption that all energy use is on-grid. The impact of off-grid renewable sources use is not negligible and we encourage its consideration. If both Lightning throughput and up-to-date off-grid mining data are considered, Bitcoin's carbon footprint per transaction amounts to only a quarter of Trinomics' estimate, which significantly impacts results. We strongly advocate reliance on up-to-date and off-grid-inclusive data sources.

A proper understanding of the technology

Finally, reason for concern emerged from the fact that the Spanish energy outage was attributed to energy scarcity, which in turn was described as an issue aggravated by Bitcoin. Several attendees considered this to be an unsubstantiated statement, as there is no evidence connecting the two, but the impression was given that this connection was pristine. Arguably, the opposite could be true: The Spanish blackout was caused by transmission failures, not consumption overload. Mining, particularly in Europe, is known to be able to mitigate node congestion due to demand-response and buildout incentive. As a result, there is some concern that mining incentives are being fundamentally misunderstood in this context.

This can be further extended to blockchain technology in general. At the webinar, the statement was made that "crypto requires ongoing asset creation (e.g., mining), unlike conventional systems". This is not correct:

- Most crypto asset systems do not have mining.
- Mining is not about asset creation, but about transaction validation, and can exist without asset creation.
- The equivalent to mining in PoW's main alternative (PoS), staking, is also not about asset creation. Ethereum, in fact, is the main PoS network at present, uses staking and is deflationary.

In sum, we acknowledge that the blockchain ecosystem is very complex, and generalisations are unavoidable. In this context, however, we encourage nuance and the avoidance of sweeping statements. While blockchain technology has antifragile tendencies, it is likely that any regulation – or research that informs regulation – that is not based in an integral understanding of the technology could backfire, hurt the development of the sector in the EU and, by extension, its competitiveness,

As one of the few initiatives working closely with regulators, CASPs, and crypto asset projects, the MiCA Crypto Alliance has developed a practical understanding of how compliance and sustainability standards work on the ground. We have helped leading projects prepare MiCA compliant white papers and ESG disclosures, using methodologies independently validated by institutions like the University of Cambridge. We would be happy to stay engaged as the study progresses and to support the finalisation of your recommendations with field-tested insight and implementation experience.

MiCA Crypto Alliance

The MiCA Crypto Alliance is a leading collaborative initiative simplifying regulatory compliance across the crypto industry. We provide verified sustainability data and write MiCA-compliant white papers to help token issuers, CASPs and crypto projects meet their disclosure obligations under MiCA.

This alliance focuses on standardising compliance efforts among its members, offering exclusive resources like sustainability indicators and white paper elaboration tools tailored to meet MiCA requirements. By leveraging the collective expertise of its members, the MiCA Crypto Alliance will help reduce the complexities and costs associated with compliance, while setting a high standard for transparency, market integrity, and consumer protection. For more details on joining the MiCA Crypto Alliance.

Visit: micacryptoalliance.com

Contact us: mica@dltscience.org